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RTCA/DO-160G

Environmental Conditions and Test Procedures for Airborne Equipment

User's Guide Appendix 18

Audio Frequency Conducted Susceptibility – Power Inputs

Based on
DO-160G Section 18.

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User's Guide, Appendix 18

Section 18 defines the environment, test procedures and requirements. In order to properly interpret the procedures and requirements outlined in Section 18, a User's Guide is provided in Appendix 18.

The User's Guide includes rationale, guidance and background information for the environment, test procedures and requirements, guidance in applying the requirements, and lessons learned from aircraft and laboratory experience. This information should help user's understand the intent behind the requirements, should aid in tailoring requirements as necessary for particular applications and should help user's develop detailed test procedures based on the general test procedures in Section 18.

The User's Guide follows a parallel format to the main body of Section 18 for easy cross reference. Paragraph numbers corresponding to each paragraph in Section 18 are included the User's Guide. Where there is no additional information provided in the User's Guide, the paragraph header is provided but the following paragraph is left blank.

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18.A.0 Audio Frequency Conducted Susceptibility - Power Inputs

18.A.1 Purpose of the Test

This section of DO-160 is intended to provide test procedures and test levels that can be used to test equipment for audio frequency conducted susceptibility of power input lines. It is the responsibility of the installer to make sure the test results satisfy the certification requirements of the proposed installation. This user's guide does not contain requirements; it is intended to provide background information and considerations that improve the likelihood of successful test results.

18.A.2 Equipment Categories and Frequency Classes

For AC equipment, Categories R(CF), R(NF), and R(WF) correspond to Categories A(CF), A(NF), and A(WF) in section 16, respectively.

For DC equipment, Categories R, B, and Z correspond to Categories A, B, and Z in section 16, respectively.

18.A.2.1 Equipment Categories

18.A.3 Test Procedures

a. In some installations, the power return is not connected locally to the ground plane, but rather, is isolated from ground. In these cases, it may become necessary to isolate the voltage measurement device from ground in order to properly measure the injected signal. Several methods exist to isolate the voltage measurement device, including the use of differential probe amplifiers and isolation transformers inline with the power input to the measurement device. The benefit to the use of a differential probe amplifier is that it does not present a potential shock hazard, as does the use of an isolation transformer on the power input to the measurement device.

18.A.3.1 DC Input Power Leads

18.A.3.2 AC Input Power Leads

a. Phase shifting networks (PSN) should generally be avoided when performing this test. PSNs introduce extra elements in the measurement path, and can introduce errors in measurement resulting in either overtest or undertest, if not tuned properly.

b. The figures in section 18 show all measurements being taken across the EUT **INPUT TERMINALS**. While the capacitor across the power supply serves to provide a low impedance for the AFCS test signal, so that most of the AFCS signal delivered to the system is applied to the EUT, there is still an impedance associated with the supply and capacitor. Kirchoff's voltage law states that the voltage input across the secondary winding of the isolation transformer will be equal to the sum of the voltages across the EUT and the capacitor/power supply combination. This results in the voltage across the EUT being lower than the voltage across the secondary winding of the isolation transformer. For this reason, measurements should not be taken across either the secondary winding of the transformer, or "voltage measurement windings", if present, on the transformer.

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18.A.3.3 Frequency Scan Rates

The test steps per decade and dwell times were determined as the optimum values based on "best" designs and "worst" designs. In the case of a "worst" design, more steps are needed to find all of the failure points. In the case of "best" designs, the longer dwell times are needed to induce overheating in damping elements of the input power circuitry.

18.A.4 General Remarks

a. The 36 Amp (peak-to-peak) current limit is based on the 100Wrms power limit and 0.6 Ohm rated impedance present in DO-160E. Both originated from test equipment historically used during performance of this test. 36Ap-p is derived form 100Wrms as follows:

$P_{rms} = (I_{rms}\text{^}2)*Z$
$I_{p\text{-}p} = I_{rms} * (\text{sqrt } 2) * 2$

$100W_{rms}$, across 0.6 ohms: sqrt (100W / 0.6 ohm)= $12.91A_{rms}$ = 36.52 amp $_{pk\text{-}pk}$

b. The 0.6 ohm rated impedance is based on legacy equipment used during the early versions of this test. The tube amplifiers used when these specifications were standardized had roughly a 2 – 2.5 ohm output impedance, which resulted in a 0.5 – 0.6 ohm rated load impedance on the secondary winding of the isolation transformer.

c. When a transformer is used to couple the audio frequency energy into the power lead, it must be capable of performing adequately when the ac or dc load current drawn by the equipment under test flows through its secondary winding.

d. On AC lines, phase shifting networks should not be used, due to the potential for inaccurate measurements when using these devices.

e. Caution must be exercised so that reflected voltages developed by input power current do not damage the audio power source generation system.

f. The dc offset of the audio generator feeding the transformer can cause the audio generator and/or transformer to overheat and fail. A capacitor paralleled with a 1 Ohm resistor inserted between the audio generator and transformer will prevent this problem. Both the capacitor and resistor must be rated for the current passing through it. The capacitor must be low enough in impedance to allow the required test levels to be achieved.

Figure 18-A-1 Test Setup for Audio Frequency Conducted Susceptibility Test (For AC and DC Power Lines, Differential Mode)

a. The current monitor is listed as optional to allow test conductors to only monitor test voltage during test, despite the risk for over test. The risk in this practice is that the possibility exists of the EUT input filtering becoming resonant at the AFCS signal frequency, and requiring currents greater than 36Ap-p to reach the desired test voltage, resulting in potential damage to the EUT.

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Figure 18-A-2 Test Setup for Audio Frequency Conducted Susceptibility Test (For 270V DC Power Lines, Common Mode)

a. The current monitor is listed as optional to allow test conductors to only monitor test voltage during test, despite the risk for over test. The risk in this practice is that the possibility exists of the EUT input filtering becoming resonant at the AFCS signal frequency, and requiring currents greater than 36Ap-p to reach the desired test voltage, resulting in potential damage to the EUT.

Figure 18-A-3 Frequency Characteristics of Ripple in 28 & 270 Volt DC Electric System — Categories R, K, & Z

Figure 18-A-4 Frequency Characteristics of Ripple in 28 Volt DC Electric System — Category B

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